DATA ANALYSIS

1. **Descriptive Analysis**
   - ATM Performance & Transaction Analysis
   - Time Based & Temporal Patterns Analysis
   - Transaction Type Analysis
   - Customer Profile & Behavior
   - ATM Location & Infrastructure
   - Calendar & Seasonal Summary

2. **Diagnostic Analysis**
   - ATM underutilization Analysis
   - Customer Segment Impact on ATM usage
   - Holiday transaction behavior analysis
   - State-wise ATM demand variation analysis
   - Transaction duration root cause analysis
   - Wisabi vs Non-Wisabi customer behavior analysis

1. Why ATM Performance & Transaction Analysis Matters

This analysis is a performance monitoring and capacity optimization system for ATM network operations in Nigeria.

It answers three core banking questions:

- Where is demand highest?
- Where is revenue generated most?
- Where are machine underutilized or overloaded?

This analysis helps:

- Identify high and low performing ATM regions.
- Optimize ATM allocation
- Improve revenue generation.
- Detect growth patterns
- Support strategic expansion
- Reduce operational inefficiencies

Target Audience:

1. **Head of ATM Operations / ATM Network Manager**

By analyzing transaction volume per ATM, and regional transaction trends, we can optimize ATM allocation, reduce machine overload, prevent cash-out situations, and improve reliability without increasing infrastructure costs.

It helps them to:

- Identifies overloaded ATMs
- Detects cash replenishment planning
- Supports downtime reduction

Benefits they get:

- Lower operational cost
- Better machine utilization
- Reduced customers complaints
- Improved ATM uptime
- Higher efficiency per ATM

2. **Regional Business Heads**

By evaluating transaction performance across states and cities, along with growth trends over time, we can identify high-performing regions, detect emerging markets, and make data-driven expansion decisions.

How it helps them

- Shows which region drives the most revenue

- Highlights fastest-growing states
- Reveals seasonal demand patterns
- Supports regional performance benchmarking

Benefits They get:

- Smarter expansion planning
- Focused regional investment
- Higher regional profitability.
- Early identification of declining markets.

3. Finance Department

By analyzing total transaction value, average transaction size and regional contribution trends, we can improve revenue tracking, enhance ROI analysis, and support better financial planning for ATM infrastructure.

How it helps them

- Estimates revenue contribution by region.
- Assesses ROI per ATM
- Tracks yearly financial growth
- Detects unusual transaction patterns

Benefits They get

- Accurate budget allocation
- Improved profitability analysis
- Better capital utilization
- Reduced financial risk

4. Strategy & Expansion Team

By studying month-over-month, quarter-over-quarter and year-over-year growth patterns, we can forecast demand, identify expansion opportunities, and optimize geographic ATM coverage.

How it helps them:

- Identifies high-growth cities.
- Detects saturation zones
- Supports long-term planning
- Enables demand forecasting

Benefits they get

- Higher return on expansion investment

- Reduced infrastructure risk
- Data-backed strategic decisions
- Sustainable network growth

Conclusion:

This analysis transforms raw transaction data into actionable insights that improve operational efficiency, enhance customer experience, optimize infrastructure investment, and support strategic growth decisions across the ATM network.

2. Why Time-Based & Temporal Pattern Analysis Matters

The above analysis is focused on Time-Based performance & Demand Optimization Framework for ATM network operations in Nigeria.

It studies how transaction behaviour changes across:

- Hours of the day
- Days of the week
- Weeks of the year
- Months & Quarters
- Holidays and pre/post-holidays periods

The primary objective of this analysis is To optimize ATM availability, cash management, and operational efficiency by understanding temporal transaction patterns.

Target Audience & Business Impact.

1. Head of ATM Operations / ATM Network Manager

By analyzing transaction volume by hour, day, week, and holiday periods, we can optimize scheduling, reduce overload, and improve machine reliability.

How it helps them:

- Identifies peak and off-peak hours
- Detects congestion periods
- Highlights pre-holiday demand surges
- Supports better refill timing
- Identifies slow operational windows for maintenance

Benefits They Get:

- Reduce machine overload
- Fewer cash-out incidents

- Lower emergency refill costs
- Improved ATM uptime
- Better service reliability
- Reduced customer complaints

2. Regional Business Heads

By comparing peak hours and demand cycles across regions, we can identify behavioral differences and regional demand intensity.

How it helps them:

- Reveals regional peak demand hours
- Highlights seasonal demand spikes
- Shows weekend vs weekday behavior
- Identifies region-specific usage patterns

Benefits They get:

- Better regional planning
- Smarter resource allocation
- Improved service reliability per region.
- Data-driven regional performance evaluation.

3. Finance Department:

By analyzing seasonal spikes, quarterly transaction cycles, and demand fluctuations, we improve finanical planning and cost forecasting.

How it helps them

- Tracks quarterly demand cycles
- Forecasts high-cash requirement periods
- Identifies low-utilization periods
- Detects abnormal seasonal spikes

Benefits They Get:

- Improved cash flow planning
- Better budgeting accuracy
- Optimized working capital
- Reduced operational waste

4. Strategy & Expansion Team

By studying long-term time patterns and seasonal growth trends, we can forecast demand and expansion planning

How it helps them

- Identifies growing seasonal demand
- Detects consistent year-round high utilization
- Reveals long-term demand stability
- Supports demand forecasting models.

Benefits they get

- Lower infrastructure risk.
- Smarter geographic scaling
- Sustainable ATM network growth

Conclusion:

This Time-Based & Temporal Pattern Analysis transforms transaction timestamps into actionable operational intelligence.

It enables:

- Pro active ATM management
- Efficient resource allocation
- Reduced operational cost
- Improved customer experience
- Data-driven strategic planning

3.Why Transaction Type Analysis Matters

This analysis focuses on understanding ATM transaction behavior based on different transaction types such as:

- Withdrawal
- Deposit
- Balance inquiry
- Fund Transfer

By analyzing:

- Total number of transactions
- Total transaction value
- Region-wise usage
- Hourly usage pattern
- Monthly transaction trends
- Average transaction amount
- Withdrawal-to-deposit ration per ATM location

The objective of this analysis is to evaluate how customers use ATM services across time, location, and transaction categories, and how it impacts the banks cash flow and ATM operations.

Why I did the analysis

Different ATM transactions impact the bank differently:

Withdrawal generates Cash outflow in bank
Deposit generate cash inflow in bank
Balance Inquiry creates Network cost for bank
Fund Transfer creates Fee income for bank

So bank need to understand

- Which transaction types dominate ATM usage
- Which regions are withdrawal-heavy or deposit-heavy
- When ATMs are used the most during the day
- Monthly trends in cash withdrawal and deposit
- Average transaction size per transaction type
- ATM locations with high cash outflow pressure

So, this analysis helps the bank manage ATM liquidity efficiently and reduce operational inefficiencies.

Target audience and benefits

- **ATM operations Team**
  This analysis helps them identify high-usage ATMs, peak transaction hours, and withdrawal-heavy locations.
  Benefits:
  - Reduce ATM downtime
  - Improves cash replenishment scheduling
  - Enhances service availability and customer satisfaction
- **Treasury / Cash Management Team**
  By understanding withdrawal-to-deposit trends and monthly transaction patterns, they can forecast cash demand more accurately.
  Benefits:
  - Better liquidity management
  - Reduced idle cash in ATMs
  - Lower cash-in-transit and replenishment costs

- **Regional Business Heads**
  Region-wise transaction behavior helps them evaluate ATM performance across different states.
  Benefit:

- Improved regional planning
- Identification of high-demand areas
- Data-driven decision-making for ATM expansion or optimization

- **Strategy & Planning Team**
  Insights into transaction distribution and trends support long-term infrastructure decisions.
  Benefits:

- Optimized ATM network deployment
- Better capital allocation
- Informed decisions on installing deposit-enabled ATMs

- **Risk Monitoring Team**

  Monitoring transaction value trends and abnormal spikes supports fraud detection efforts.

  Benefits

  – Early identification of unusual transaction patterns
  – Strengthened risk controls
  – Reduced financial exposure.

4.Why Customer Profile & Behavior Matters?

The above analysis helps to understand who is using our ATM network, how they are using it, and which type of customers are contributing the most to ATM transaction volume and value.
It segments customers based on:

1. Transaction amount
2. Transaction frequency
3. Age group
4. Gender
5. Occupation
6. Account type
7. Location
8. Product ownership (Wisabi VS Non-Wisabi)

This analysis is better to

- Identify which customers generate maximum ATM usage
- Improve ATM network utilization
- Increase customer engagement
- Encourage internal ATM usage (Wisabi Customers)
- Support targeted marketing & product strategry

Target Audience & Benefits

1. **Retail Banking Team**
   They can easily find who are our most valuable ATM customers.
   And which customers should be retained.
   So they can easily offer premium banking, Pre-approved loans, Account upgrades
2. **ATM Operations Team**

Through this analysis they can get which customer segments use ATM more?

And Which locations serve high-value customers?
So they can easily

- Oprimize ATM placement
- Plan cash replenishment

- Avoid machine overload
- Improve service uptime

3. **Marketing Team**
   This analysis helps marketing team to know
   Which age group transacts more?
   Which occupation uses ATMs more?
   Do Wisabi customers transact more?
   So they can run targeted campaigns, Promote ATM – linked products, Increase Wisabi onboarding and Cross-sell credit cards or loans
4. **Product Team**
   Through this analysis Product team will get are Wisabi customers more engaged than non-wisabi?
   So they can improve product adoption, increase internal ATM usage and Reduce third party ATM dependency.

5. Why ATM Location & Infrastructure Matters

This analysis helps the bank evaluate the performance and utilization of its ATM infrastructure across locations, enabling data-driven decisions for ATM placement, cost optimization, revenue improvement, and network expansion planning.

- ATM availability across cities & states
- Reduced financial exposure.
- Transaction load handled by each ATM
- Customer usage across locations
- Country-wise ATM presence

So, the bank can clearly identify:

- High performing ATM locations
- Underutilized ATM locations
- Locations with high customer demand
- Locations with infrastructure gaps

Target Audience & Benefits

1. **ATM Operations Team**
   This analysis helps them to know:
   - Which ATM locations handle more transactions?
   - Which locations have higher customer usage?

   So, they can improve cash replenishment schedules, Avoid machine overload Reduce downtime Allocate maintenance resources efficiently.

2. **Infrastructure Planning Team**
   This analysis explains
   ATM distribution by city/state
   Customer load per location
   So, they can install ATMs in high-demand areas, Remove or relocate underutilized machines
   Plan future ATM expansion
3. **Finance Team**
   This analysis helps Finance team to know
   - Revenue generated per location
   - Average transaction value per ATM

   So, they can evaluate ROI on ATM installations, Control operational costs, Prioritize investment in profitable locations

6. Why Calendar & Seasonal Summary Matters?

This analysis helps the bank understand seasonal ATM transaction patterns across holidays, months, and quarters, enabling proactive cash management, improved ATM uptime, and data-driven operational planning.

Target audience and benefits

1. ATM Operations Team

This analysis helps them to find

- When transaction demand is highest
- Do holidays increase ATM usage?

  So they can Load additional cash before peak days, prevent ATM outages and improve uptime during holidays.

2. Cash Management Team

This analysis helps them to know

- Monthly average transaction load
- Peak transaction dates.

  So they can plan efficient cash logistics, Avoid emergency replenishments and Reduce idle cash storage cost.

3. Strategy & Planning Team

This analysis helps them to know

- Quarterly performance trends

  So they can align ATM network planning, Anticipate seasonal usage and improve infrastructure readiness.

4. Customer Experience Team

This analysis helps them to know

- Holiday usage patterns

  So they can ensure ATM availability. Reduce transaction failures, improve service satisfaction.

DIAGNOSTIC ANALYSIS

1. Why ATM underutilization Analysis matters?

Some ATM locations show low transactions volume despite having a high number of ATMs installed. This leads to inefficient infrastructure investment and increased maintenance costs.

This analysis identifies whether low usage is caused by:

- Fewer unique customers
- Limited transaction type usage
- Time-specific demand (peak-hour only usage)
- Holiday or weekend inactivity

Target Audience and their benefits:

This analysis is helpful for the **ATM Network Planning Team** to identify locations where ATM machines are underutilized despite having higher infrastructure. This enables them to detect redundant ATM installations and take data-driven decisions regarding relocation or redistribution of ATM machines.

Additionally, the **Infrastructure Management Team** benefits from this analysis by avoiding unnecessary ATM deployment in low-demand areas. It also helps them plan ATM availability based on time-specific demand patterns, ensuring that ATM resources are efficiently utilized during peak usage hours while minimizing idle infrastructure during low-demand periods.

2. Why Customer Segment Impact on ATM usage analysis matter?

Different customer groups interact with ATM services differently, understanding these behavioral differences helps explain low usage in certain locations.

This analysis determines

- Which account types rely less on ATMs
- Which occupations prefer ATM usage
- Whether Wisabi users behave differently.
- Which customer segments contribute least to ATM usage

Target Audience & Their Benefits:

This analysis supports the **Customer Insights Team** in understanding how different customer segments such as account types, occupations, or Wisabi users interact with ATM services. It helps identify customer groups that are less engaged with ATM usage.

The **Product Management Team** can leverage these insights to design targeted engagement strategies or customized banking services for underutilizing customer segments. This ultimately improves ATM adoption among specific groups and enhances overall customer experience.

3. Why does Holiday transaction behavior analysis matter

ATM usage often fluctuates during holidays, identifying these variations helps understand transaction spikes or drops

This analysis identifies

- Which transaction increase during holidays
- Whether Wisabi users transact more during holidays
- Locations with holiday-driven spikes
- Customer segments responsible for holiday usage

Target Audience & Their Benefits:

This analysis is valuable for the **ATM Cash Management Team**, as it enables them to anticipate transaction spikes during holidays and ensure sufficient cash loading in high-demand ATM locations.

The **Operations Planning Team** can use these insights to proactively manage service availability by scheduling maintenance activities during low-demand periods and preventing ATM downtime during peak festive usage. This improves service reliability and customer satisfaction during critical time periods.

4. Why does State-wise ATM demand variation analysis matters

ATM usage patterns differ across states due to regional bahavior, economic activity, and payment preferences.

This analysis evaluates:

- ATM demand across states
- Withdrawal- heavy regions
- Weekend transaction trends
- Holiday usage variatioin by state

Target Audience & Their Benefits:

This analysis assists the **Regional Banking Managers** in understanding how ATM demand varies across different states based on regional transaction behavior.

It also supports the **Expansion Strategy Team** in making informed decisions about region-specific ATM deployment. By identifying high-demand states, the team can prioritize infrastructure expansion in growth areas and ensure efficient allocation of ATM resources across regions.

5. Why does Transaction duration root cause analysis matters

Long transaction duration may indicate ATM congestion, network latency, or customer familiarity issues.

This analysis identifies

- Locations with longer transaction durations
- Peak-hour congestion
- Holiday transaction delays
- Occuapation-based usage time differences

Target Audience & Their Benefits:

This analysis benefits the **ATM Performance Monitoring Team** by identifying locations where transaction completion times are consistently high, indicating possible congestion or system performance issues.

The **IT Support Team** can utilize this information to investigate potential network latency or software-related delays during peak hours or holidays. This leads to improved system performance, reduced customer wait time, and enhanced ATM service efficiency.

6. Why does Wisabi vs Non-Wisabi customer behavior analysis matter

Wisabi customers may have different usage patterns due to enhanced banking services.

This analysis evaluates

- Preferred transactin types
- Weekday vs weekend usage
- Hour-wise transaction behavior
- Location preference of Wisabi users

Target Audience & Their Benefits:

This analysis is useful for the **Premium Banking Team** to understand the ATM usage behavior of Wisabi customers compared to Non-Wisabi users.

The **Customer Experience Team** can leverage these insights to tailor ATM services or prioritize availability in locations frequently accessed by Wisabi users. This enables the bank to deliver differentiated services and improve satisfaction among premium customer segments.

1. How many distinct City are available in Nigeria

```
SELECT COUNT(DISTINCT(City)) City_Count
FROM [atm_location ]
WHERE Country = 'Nigeria'
```

	City_Count
1	19

2. . How many states are available in Nigeria

```
SELECT DISTINCT(State) AS States
FROM [atm_location ]
WHERE Country= 'Nigeria'
```

	States
1	Enugu
2	Federal Capital Territory
3	Kano
4	Lagos
5	Rivers State

ATM Performance & Transaction Analysis

- Total transactions and total value by region (State)

```
SELECT t1.State,
  COUNT(DISTINCT t2.TransactionID) as Total_Transactions,
  SUM(CAST(t2.TransactionAmount as BIGINT)) as Total_Amount
FROM [atm_location ] as t1
INNER JOIN fact_transactions as t2
ON t1.LocationID = t2.LocationID
GROUP BY t1.State
ORDER BY Total_Transactions DESC,Total_Amount DESC;
```

	State	Total_Transactions	Total_Amount
1	Lagos	755073	14232248000
2	Kano	458764	7989721000
3	Rivers State	420098	7352145000
4	Enugu	350251	6135100000
5	Federal Capital Territory	159652	2846671000

- Calculate Total transactions and value by City

```
SELECT t1.City,
        COUNT(DISTINCT t2.TransactionID) as Total_Transactions,
```

```
        SUM(CAST(t2.TransactionAmount as BIGINT)) as Total_Amount
FROM [atm_location ] as t1
INNER JOIN fact_transactions as t2
ON t1.LocationID = t2.LocationID
GROUP BY t1.City
ORDER BY Total_Transactions DESC,Total_Amount DESC
```

	City	Total_Transactions	Total_Amount
1	Lagos	755073	14232248000
2	Kano	458764	7989721000
3	Enugu	159973	2801533000
4	Abuja	159652	2846671000
5	Port Harcourt	48954	861191000
6	Eleme	45993	807200000
7	Opobo	45552	794208000
8	Gokana	44566	775885000
9	Udi	43795	763474000
10	Emohua	43365	750466000
11	Okrika	40612	710827000

11	Okrika	40612	710827000
12	Oji	40376	703950000
13	Ahoada	39881	708994000
14	Nkanu	38865	683153000
15	Bonny	38035	665645000
16	Oyigbo	37804	660081000
17	Obio-Akpor	35336	617648000
18	Awgu	34074	596371000
19	Nsukka	33168	586619000

➢ Calculate Total transactions and Value by ATM Location/Branch

```
SELECT
   t1.Location_Name,
   COUNT(t2.TransactionID) as Total_Transactions,
   SUM(t2.TransactionAmount) as Total_Amount
FROM atm_location t1
LEFT JOIN fact_transactions t2
ON t1.LocationID = t2.LocationID
GROUP BY t1.Location_Name
ORDER BY Total_Transactions DESC, Total_Amount DESC
```

	Location_Name	Total_Transactions	Total_Amount
1	Lekki ATM	55465	1041469000
2	Sabon Gari ATM	52823	919076000
3	Ikorodu ATM	51754	973858000
4	Surulere ATM	51533	978496000
5	Ajah ATM	51228	961600000
6	Oshodi ATM	51076	967518000
7	Yaba ATM	51044	955450000
8	Festac ATM	50418	957593000

	Location_Name	Total_Transactions	Total_Amount
9	Tarauni ATM	50381	883299000
10	Ikeja ATM	49643	940147000
11	Ikoyi ATM	49633	933887000
12	Apapa ATM	49254	923069000
13	Wisabi Bank HQ	49130	915894000
14	Ogba ATM	48989	930370000
15	Port Harcourt ...	48954	861191000
16	Victoria Island ...	48787	921525000

➢ Calculate Average Transaction Value per Location

```
SELECT
   t1.Location_Name,
```

```
    COUNT(t2.TransactionID) as Total_Transactions,
    AVG(t2.TransactionAmount) as Avg_Amount
FROM atm_location t1
LEFT JOIN fact_transactions t2
    ON t1.LocationID = t2.LocationID
GROUP BY t1.Location_Name
ORDER BY Total_Transactions DESC, Avg_Amount DESC
```

	Location_Name	Total_Transactions	Avg_Amount
1	Lekki ATM	55465	18777
2	Sabon Gari ATM	52823	17399
3	Ikorodu ATM	51754	18817
4	Surulere ATM	51533	18987
5	Ajah ATM	51228	18770
6	Oshodi ATM	51076	18942
7	Yaba ATM	51044	18718
8	Festac ATM	50418	18993
9	Tarauni ATM	50381	17532
10	Ikeja ATM	49643	18938
11	Ikoyi ATM	49633	18815

	Location_Name	Total_Transactions	Avg_Amount
11	Ikoyi ATM	49633	18815
12	Apapa ATM	49254	18740
13	Wisabi Bank HQ	49130	18642
14	Ogba ATM	48989	18991
15	Port Harcourt ...	48954	17591
16	Victoria Island ...	48787	18888
17	Maryland ATM	48694	18870
18	Nassarawa ATM	48527	17507
19	Agege ATM	48425	18843
20	Kofar Ruwa A...	47044	17251
21	Eleme ATM	45993	17550

➢ Find out Transactions per ATM Machine

```
SELECT
        t1.LocationID,
        t1.State,
        t1.City,
        t1.No_of_ATMs,
COUNT(t2.TransactionID) AS Total_Transactions,
ROUND((COUNT(t2.TransactionID) * 1.0 / t1.No_of_ATMs),2) AS
Transactions_per_ATM
    FROM [atm_location] AS t1
    LEFT JOIN fact_transactions AS t2
    ON t1.LocationID = t2.LocationID
   GROUP BY  t1.LocationID,
      t1.State,
      t1.City,
      t1.No_of_ATMs
  ORDER BY Transactions_per_ATM DESC
```

	LocationID	State	City	No_of_ATMs	Total_Transactions	Transactions_per_ATM
1	KN-002	Kano	Kano	2	52823	26411.500000
2	KN-006	Kano	Kano	2	50381	25190.500000
3	KN-004	Kano	Kano	2	48527	24263.500000
4	KN-005	Kano	Kano	2	47044	23522.000000
5	RI-005	Rivers State	Eleme	2	45993	22996.500000
6	RI-010	Rivers State	Opobo	2	45552	22776.000000
7	KN-007	Kano	Kano	2	44985	22492.500000
8	RI-007	Rivers State	Gokana	2	44566	22283.000000
9	KN-010	Kano	Kano	2	44055	22027.500000
10	EN-007	Enugu	Udi	2	43795	21897.500000
11	RI-006	Rivers State	Emohua	2	43365	21682.500000

- Month-over-MOnth and Quarter – over- Quarter transaction trends Growth or Decline patterns over time

  -- Month-over-Month transaction trend

```
SELECT  t1.Year,
        t1.Month,
        t1.Month_Name,
        COUNT(t2.TransactionID) AS Total_Transactions
FROM calendar AS t1
INNER JOIN fact_transactions AS t2
ON t1.Date = t2.DateKey
GROUP BY  t1.Year,
          t1.Month,
          t1.Month_Name
ORDER BY  t1.Year,
          t1.Month;
```

	Year	Month	Month_Name	Total_Transactions
1	2022	1	January	178913
2	2022	2	February	161418
3	2022	3	March	196327
4	2022	4	April	172953
5	2022	5	May	183905
6	2022	6	June	173217
7	2022	7	July	184362
8	2022	8	August	178712
9	2022	9	September	173443
10	2022	10	October	184164
11	2022	11	November	172335

-- Quarter-over-Quarter transaction trend

```
SELECT t1.Year, t1.Quarter,
        COUNT(t2.TransactionID) AS Total_Transactions
FROM [calendar] AS t1
INNER JOIN fact_transactions AS t2
ON t1.Date = t2.DateKey
GROUP BY t1.Year, t1.Quarter
ORDER BY t1.Year, t1.Quarter;
```

	Year	Quarter	Total_Transactions
1	2022	Q1	536658
2	2022	Q2	530075
3	2022	Q3	536517
4	2022	Q4	540588

➢ Year-on-Year comparison of transactions volume and value.

```
SELECT t1.Year,
   SUM(CAST(t2.TransactionAmount AS BIGINT)) Total_Amount,
   AVG(CAST(t2.TransactionAmount AS BIGINT)) Avg_Amount,
   MIN(CAST(t2.TransactionAmount AS BIGINT)) Min_Amount,
   MAX(CAST(t2.TransactionAmount AS BIGINT)) Max_Amount
FROM [calendar ] t1
INNER JOIN fact_transactions AS t2
ON t1.Date = t2.DateKey
GROUP BY t1.Year
ORDER BY Total_Amount DESC;
```

	Year	Total_Amount	Avg_Amount	Min_Amount	Max_Amount
1	2022	38555885000	17984	0	100000

➢ Cumulative transactions value by region (running total)

Contribution pace across the year.

```
SELECT t1.State,
```

```
    t2.transactionStartDateTime,
SUM(CAST(t2.TransactionAmount AS BIGINT))
OVER(PARTITION BY t1.State ORDER BY t2.transactionStartDateTime) as Cumulative_Amount
FROM [atm_location ] AS t1
INNER JOIN fact_transactions AS t2
ON t1.LocationID = t2.LocationID
ORDER BY t1.State, t2.transactionStartDateTime;
```

	State	transactionStartDateTime	Cumulative_Amount
1	Enugu	2022-01-01 07:01:00.0000000	3000
2	Enugu	2022-01-01 07:02:00.0000000	16000
3	Enugu	2022-01-01 07:02:00.0000000	16000
4	Enugu	2022-01-01 07:06:00.0000000	36000
5	Enugu	2022-01-01 07:06:00.0000000	36000
6	Enugu	2022-01-01 07:07:00.0000000	46000
7	Enugu	2022-01-01 07:10:00.0000000	46000
8	Enugu	2022-01-01 07:11:00.0000000	66000
9	Enugu	2022-01-01 07:14:00.0000000	75000
10	Enugu	2022-01-01 07:15:00.0000000	115000
11	Enugu	2022-01-01 07:15:00.0000000	115000

	State	transactionStartDateTime	Cumulative_Amount
13	Enugu	2022-01-01 07:21:00.0000000	132000
14	Enugu	2022-01-01 07:23:00.0000000	143000
15	Enugu	2022-01-01 07:23:00.0000000	143000
16	Enugu	2022-01-01 07:24:00.0000000	213000
17	Enugu	2022-01-01 07:24:00.0000000	213000
18	Enugu	2022-01-01 07:24:00.0000000	213000
19	Enugu	2022-01-01 07:25:00.0000000	230000
20	Enugu	2022-01-01 07:25:00.0000000	230000
21	Enugu	2022-01-01 07:27:00.0000000	250000
22	Enugu	2022-01-01 07:28:00.0000000	261000
23	Enugu	2022-01-01 07:28:00.0000000	261000

Time Based & Temporal Patterns

Analyzing how data changes over time to identify trends, seasonality, cycles, or unusual behavior.

➢ Transaction volume by hour of day

```
SELECT
   t1.Hour_Key,
   COUNT(t2.TransactionID) as Total_Transactions
FROM hours_table t1
LEFT JOIN fact_transactions t2
   ON t1.Hour_Key = t2.Hour_Key
GROUP BY t1.Hour_Key
ORDER BY t1.Hour_Key;
```

	Hour_Key	Total_Transactions
1	0	4
2	1	4
3	2	5
4	3	3
5	4	1333
6	5	4507
7	6	4658
8	7	115903
9	8	114313
10	9	114930
11	10	114651

14	13	135472
15	14	134501
16	15	198773
17	16	233349
18	17	233890
19	18	195189
20	19	58921
21	20	53157
22	21	53354
23	22	53182
24	23	53437

➢ Transaction volume by day of week

```
SELECT t1.Day_of_Week,
    t1.Day_name,
    COUNT(t2.TransactionID) as Total_Transactions
FROM [calendar ] AS t1
INNER JOIN fact_transactions as t2
ON t1.Date = t2.DateKey
GROUP BY t1.Day_of_Week,
 t1.Day_Name
ORDER BY t1.Day_of_Week
```

	Day_of_Week	Day_name	Total_Transactions
1	0	Monday	299710
2	1	Tuesday	305981
3	2	Wednesday	304367
4	3	Thursday	312123
5	4	Friday	304828
6	5	Saturday	311606
7	6	Sunday	305223

➢ Transaction Volume by week of year

```
SELECT t1.Year,
        t1.Week_of_Year,
        COUNT(t2.TransactionID) AS TransactionVolume
FROM calendar t1
INNER JOIN fact_transactions t2
  ON t1.Date = t2.DateKey
GROUP BY t1.Year,
            t1.Week_of_Year
ORDER BY t1.Year,
            t1.Week_of_Year;
```

	Year	Week_of_Year	TransactionVolume
1	2022	1	11436
2	2022	2	40292
3	2022	3	40433
4	2022	4	40581
5	2022	5	40500
6	2022	6	40353
7	2022	7	40347
8	2022	8	40177
9	2022	9	40428
10	2022	10	57854
11	2022	11	40343

	Year	Week_of_Year	TransactionVolume
12	2022	12	40434
13	2022	13	40429
14	2022	14	40343
15	2022	15	40354
16	2022	16	40089
17	2022	17	40571
18	2022	18	46191
19	2022	19	40208
20	2022	20	40414
21	2022	21	40021
22	2022	22	40250

➢ Transaction volume by month and quarter

-- Month wise transaction volume

```
SELECT t1.Month,
    COUNT(t2.CardholderID) as Total_Transactions
FROM [calendar ] as t1
INNER JOIN fact_transactions as t2
ON t1.Date = t2.DateKey
GROUP BY t1.Month
```

	Month	Total_Transactions
1	1	178913
2	2	161418
3	3	196327
4	4	172953
5	5	183905
6	6	173217
7	7	184362
8	8	178712
8	8	178712
9	9	173443
10	10	184164
11	11	172335
12	12	184089

-- Quarter wise Transaction Volume

```
SELECT t1.Quarter,
COUNT(t2.CardholderID) as Total_Transactions
FROM [calendar ] as t1
INNER JOIN fact_transactions as t2
ON t1.Date = t2.DateKey
GROUP BY t1.Quarter
```

	Quarter	Total_Transactions
1	Q1	536658
2	Q2	530075
3	Q3	536517
4	Q4	540588

➢ Peak hour comparison across regions

```
WITH hours_transactions AS (
    SELECT t2.State,
            DATEPART(HOUR, t1.TransactionStartDateTime) AS Hour_No,
            COUNT(t1.TransactionID) AS Total_Transactions
    FROM fact_transactions t1
    INNER JOIN atm_location t2
        ON t1.LocationID = t2.LocationID
   GROUP BY  t2.State,
                DATEPART(HOUR, t1.TransactionStartDateTime))
```

```
SELECT State, Hour_No, Total_Transactions
FROM ( SELECT  State,
               Hour_No,
               Total_Transactions,
               RANK( ) OVER (PARTITION BY State
                             ORDER BY Total_Transactions DESC) AS ranks
       FROM hours_transactions) ranked
WHERE ranks = 1;
```

	State	Hour_No	Total_Transactions
1	Enugu	17	54554
2	Federal Capital Territory	18	17072
3	Kano	11	45003
4	Lagos	15	67020
5	Rivers State	18	51130

➢ Transaction volume on holidays vs weekdays vs weekends

```
WITH DayType AS (
    SELECT t2.TransactionID,
           CASE WHEN t1.IsHoliday = 1 THEN 'Holiday'
                WHEN t1.Day_of_Week IN (6,7) THEN 'Weekend'
                ELSE 'Weekday'
           END AS Day_Type
    FROM calendar t1
    INNER JOIN fact_transactions t2
      ON t1.Date = t2.DateKey )
SELECT Day_Type,
       COUNT(TransactionID) AS Total_Transactions
FROM DayType
GROUP BY Day_Type
ORDER BY Total_Transactions DESC;
```

	Day_Type	Total_Transactions
1	Weekday	1804419
2	Weekend	293658
3	Holiday	45761

- Transaction volume in the week before and after holidays

```
WITH holiday_dates AS (
        SELECT Date AS Holiday_Date
        FROM calendar
        WHERE IsHoliday = 1),

holiday_window AS (
        SELECT  c.Date,
                h.Holiday_Date,
                DATEDIFF(DAY, h.Holiday_Date, c.Date) AS Day_Diff
        FROM calendar c
        INNER JOIN holiday_dates h
          ON c.Date BETWEEN DATEADD(DAY, -7, h.Holiday_Date)
                        AND DATEADD(DAY, 7, h.Holiday_Date))

SELECT CASE WHEN Day_Diff < 0 THEN 'Before Holiday'
            WHEN Day_Diff = 0 THEN 'Holiday'
            ELSE 'After Holiday'
        END AS Period_Type,
        COUNT(t.TransactionID) AS Total_Transactions
FROM holiday_window hw
INNER JOIN fact_transactions
ON hw.Date = t.DateKey
GROUP BY CASE WHEN Day_Diff < 0 THEN 'Before Holiday'
              WHEN Day_Diff = 0 THEN 'Holiday'
              ELSE 'After Holiday'
```

```
                END
ORDER BY Period_Type;
```

	Period_Type	Total_Transactions
1	After Holiday	304690
2	Before Holiday	265095
3	Holiday	45761

- Average transaction duration(end-start) by hour and day

```
SELECT t3.Day_of_Week,
        t3.Day_Name,
        DATEPART(HOUR, t1.TransactionStartDateTime) AS Hour_No,
        AVG(DATEDIFF(SECOND,t1.TransactionStartDateTime,
                        t1.TransactionEndDateTime)) AS Avg_Duration_Seconds
FROM fact_transactions t1
INNER JOIN calendar t3
  ON t1.DateKey = t3.Date
GROUP BY t3.Day_of_Week, t3.Day_Name,
            DATEPART(HOUR, t1.TransactionStartDateTime)
ORDER BY t3.Day_of_Week, Hour_No;
```

Results | Messages

	Day_of_Week	Day_Name	Hour_No	Avg_Duration_Seconds
1	0	Monday	0	180
2	0	Monday	1	280
3	0	Monday	2	240
4	0	Monday	3	270
5	0	Monday	4	274
6	0	Monday	5	258
7	0	Monday	6	256
8	0	Monday	7	248
9	0	Monday	8	248
10	0	Monday	9	247
11	0	Monday	10	247

Transaction Type Analysis

1. Count and Value of each transaction type overall

```
SELECT  t1.TransactionTypeID,
```

```
        t2.TransactionTypeName,
        COUNT(t1.TransactionID) AS Total_Transaction_Count,
        SUM(CAST(t1.TransactionAmount AS BIGINT)) AS Total_Transaction_Value
FROM fact_transactions as t1
INNER JOIN transaction_type as t2
    ON t1.TransactionTypeID = t2.TransactionTypeID
GROUP BY t1.TransactionTypeID,   t2.TransactionTypeName
ORDER BY Total_Transaction_Value DESC;
```

	TransactionTypeID	TransactionTypeName	Total_Transaction_Count	Total_Transaction_Value
1	1	Withdrawal	1189851	25902287000
2	4	Transfer	475872	9934265000
3	2	Deposit	238850	2719333000
4	3	Balance Inquiry	239265	0

2. Transaction type distribution by region

```
SELECT t2.State AS Region,
        t1.TransactionTypeID,
        t3.TransactionTypeName,
        COUNT(t1.TransactionID) AS Total_Count,
        SUM(CAST(t1.TransactionAmount AS BIGINT)) AS Total_Value
FROM fact_transactions t1
INNER JOIN atm_location t2
  ON t1.LocationID = t2.LocationID
INNER JOIN transaction_type as t3
ON t1.TransactionTypeID = t3.TransactionTypeID
GROUP BY t2.State,
            t1.TransactionTypeID,
            t3.TransactionTypeName
ORDER BY t2.State,
            Total_Value DESC;
```

	Region	TransactionTypeID	TransactionTypeName	Total_Count	Total_Value
1	Enugu	1	Withdrawal	194440	4064528000
2	Enugu	4	Transfer	78071	1624697000
3	Enugu	2	Deposit	38823	445875000
4	Enugu	3	Balance Inquiry	38917	0
5	Federal Capital Territory	1	Withdrawal	88136	1926685000
6	Federal Capital Territory	4	Transfer	35119	744077000
7	Federal Capital Territory	2	Deposit	18387	175909000
8	Federal Capital Territory	3	Balance Inquiry	18010	0
9	Kano	1	Withdrawal	253796	5277498000
10	Kano	4	Transfer	101912	2110989000
11	Kano	2	Deposit	51438	601234000

3. Transaction type distribution by hour of day

```
SELECT DATEPART(HOUR, t1.TransactionStartDateTime) AS Hour_No,
        t2.TransactionTypeID,
        t2.TransactionTypeName,
        COUNT(t1.TransactionID) AS Total_Count,
        SUM(CAST(t1.TransactionAmount AS BIGINT)) AS Total_Value
FROM fact_transactions t1
INNER JOIN transaction_type t2
  ON t1.TransactionTypeID = t2.TransactionTypeID
GROUP BY DATEPART(HOUR, t1.TransactionStartDateTime),
        t2.TransactionTypeID,
        t2.TransactionTypeName
ORDER BY Hour_No,
        Total_Count DESC;
```

	Hour_No	TransactionTypeID	TransactionTypeName	Total_Count	Total_Value
1	0	3	Balance Inquiry	1	0
2	0	1	Withdrawal	1	30000
3	0	4	Transfer	1	20000
4	0	2	Deposit	1	5000
5	1	1	Withdrawal	2	100000
6	1	4	Transfer	1	10000
7	1	3	Balance Inquiry	1	0
8	2	1	Withdrawal	3	120000
9	2	4	Transfer	1	20000

5. Transaction type trend over months

```
SELECT YEAR(t1.TransactionStartDateTime) AS Year_No,
       MONTH(t1.TransactionStartDateTime) AS Month_No,
       DATENAME(MONTH, t1.TransactionStartDateTime) AS Month_Name,
       t2.TransactionTypeName,
       COUNT(t1.TransactionID) AS Total_Count,
       SUM(CAST(t1.TransactionAmount AS BIGINT)) AS Total_Value
FROM fact_transactions t1
INNER JOIN transaction_type t2
  ON t1.TransactionTypeID = t2.TransactionTypeID
GROUP BY YEAR(t1.TransactionStartDateTime),
         MONTH(t1.TransactionStartDateTime),
         DATENAME(MONTH, t1.TransactionStartDateTime),
         t2.TransactionTypeName
ORDER BY Year_No,
         Month_No,
         Total_Value DESC;
```

	Year_No	Month_No	Month_Name	TransactionTypeName	Total_Count	Total_Value
1	2022	1	January	Withdrawal	99006	2154004000
2	2022	1	January	Transfer	39879	830165000
3	2022	1	January	Deposit	19930	227628000
4	2022	1	January	Balance Inquiry	20098	0
5	2022	2	February	Withdrawal	89688	1955539000
6	2022	2	February	Transfer	35698	744672000
7	2022	2	February	Deposit	18132	206195000
8	2022	2	February	Balance Inquiry	17900	0
9	2022	3	March	Withdrawal	109040	2373865000
10	2022	3	March	Transfer	43448	909779000
11	2022	3	March	Deposit	22004	249554000

6. Average transaction value per transaction type

```
SELECT t2.TransactionTypeID,
        t2.TransactionTypeName,
        COUNT(t1.TransactionID) AS Total_Transactions,
        SUM(CAST(t1.TransactionAmount AS BIGINT)) AS Total_Value,
        ROUND(AVG(CAST(t1.TransactionAmount AS DECIMAL(18,2))),
                2) AS Average_Transaction_ValueROM fact_transactions t1
INNER JOIN transaction_type t2
  ON t1.TransactionTypeID = t2.TransactionTypeID
GROUP BY t2.TransactionTypeID,
          t2.TransactionTypeName
ORDER BY Average_Transaction_Value DESC;
```

	TransactionTypeID	TransactionTypeName	Total_Transactions	Total_Value	Average_Transaction_Value
1	1	Withdrawal	1189851	25902287000	21769.350000
2	4	Transfer	475872	9934265000	20875.920000
3	2	Deposit	238850	2719333000	11385.110000
4	3	Balance Inquiry	239265	0	0.000000

7. Withdrawl-to-deposit ration by location

```
SELECT l.LocationID,
        l.Location_Name,
SUM(CASE  WHEN t2.TransactionTypeName = 'Withdrawal'
          THEN 1 ELSE 0 END) AS Withdrawal_Count,
SUM(CASE WHEN t2.TransactionTypeName = 'Deposit'
THEN 1 ELSE 0 END) AS Deposit_Count,
ROUND(CAST(SUM(CASE WHEN t2.TransactionTypeName = 'Withdrawal'
THEN 1 ELSE 0 END) AS FLOAT) /
SUM(CASE WHEN t2.TransactionTypeName = 'Deposit'
THEN 1 ELSE 0 END), 2) AS Withdrawal_to_Deposit_Ratio
FROM fact_transactions t1
INNER JOIN transaction_type t2
ON t1.TransactionTypeID = t2.TransactionTypeID
INNER JOIN atm_location l
ON t1.LocationID = l.LocationID
```

```
GROUP BY l.LocationID, l.Location_Name
ORDER BY Withdrawal_to_Deposit_Ratio DESC;
```

	LocationID	Location_Name	Withdrawal_Count	Deposit_Count	Withdrawal_to_Deposit_Ratio
1	LA-015	Agege ATM	27019	5217	5.18
2	EN-001	Enugu ATM	23948	4645	5.16
3	EN-009	Emene ATM	21515	4196	5.13
4	FC-003	Garki ATM	11674	2289	5.1
5	RI-009	Ahoada ATM	22242	4370	5.09
6	RI-006	Emohua ATM	24243	4760	5.09
7	RI-004	Obio-Akpor ATM	19703	3875	5.08
8	LA-002	Victoria Island ATM	27051	5341	5.06
9	LA-001	Wisabi Bank HQ	27306	5395	5.06
10	LA-009	Yaba ATM	28343	5606	5.06
11	LA-014	Ikoyi ATM	27699	5484	5.05

8. Transaction type breakdown for Wisabi vs Non-Wisabi customers

```
WITH customer_summary AS (
  SELECT CASE
                WHEN t1.IsWisabiCustomer = 1 THEN 'Wisabi' ELSE 'Non-Wisabi'
            END AS Customer_Type,
            t2.TransactionTypeName,
            COUNT(t1.TransactionID) AS Total_Count,
            SUM(CAST(t1.TransactionAmount AS BIGINT)) AS Total_Value
  FROM fact_transactions t1
  INNER JOIN transaction_type t2
    ON t1.TransactionTypeID = t2.TransactionTypeID
  GROUP BY CASE
                WHEN t1.IsWisabiCustomer = 1 THEN 'Wisabi' ELSE 'Non-Wisabi'
                END,
                t2.TransactionTypeName)
SELECT Customer_Type,
          TransactionTypeName,
          Total_Count,
          Total_Value,
  ROUND( Total_Count * 100.0 /  SUM(Total_Count)
       OVER (PARTITION BY Customer_Type), 2) AS Volume_Share_Percent,
  ROUND( Total_Value * 100.0 / SUM(Total_Value)
       OVER (PARTITION BY Customer_Type),2) AS Value_Share_Percent
FROM customer_summary
ORDER BY Customer_Type,
              Value_Share_Percent DESC;
```

	LocationID	Location_Name	Withdrawal_Count	Deposit_Count	Withdrawal_to_Deposit_Ratio
1	LA-015	Agege ATM	27019	5217	5.18
2	EN-001	Enugu ATM	23948	4645	5.16
3	EN-009	Emene ATM	21515	4196	5.13
4	FC-003	Garki ATM	11674	2289	5.1
5	RI-009	Ahoada ATM	22242	4370	5.09
6	RI-006	Emohua ATM	24243	4760	5.09
7	RI-004	Obio-Akpor ATM	19703	3875	5.08
8	LA-002	Victoria Island ATM	27051	5341	5.06
9	LA-001	Wisabi Bank HQ	27306	5395	5.06
10	LA-009	Yaba ATM	28343	5606	5.06
11	LA-014	Ikoyi ATM	27699	5484	5.05

Customer Profile & Behavior

1. Total Unique customers using ATMs

```
SELECT
        COUNT(DISTINCT CardholderID) AS Total_Unique_Customers
FROM fact_transactions;
```

	Total_Unique_Customers
1	8819

2. Top customers by total transaction amount

```
SELECT CardholderID,
            COUNT(TransactionID) AS Total_Transactions,
            SUM(CAST(TransactionAmount AS BIGINT)) AS Total_Transaction_Value
FROM fact_transactions
GROUP BY CardholderID
ORDER BY  Total_Transaction_Value DESC;
```

	CardholderID	Total_Transactions	Total_Transaction_Value
1	EN-008-1093	846	15325000
2	FC-001-1006	672	13702000
3	FC-004-1003	665	13075000
4	FC-002-1021	742	12961000
5	FC-005-1038	703	12953000
6	FC-005-1024	684	12950000
7	FC-005-1023	655	12748000
8	FC-006-1023	694	12723000
9	FC-001-1039	684	12709000
10	FC-006-1011	645	12683000
11	FC-002-1046	632	12621000

3. State wise Top customers by total transaction amount

```
WITH State_Customer_Transactions AS (
    SELECT l.State,
                t.CardholderID,
                COUNT(t.TransactionID) AS Total_Transactions,
                SUM(CAST(t.TransactionAmount AS BIGINT)) AS Total_Transaction_Value
   FROM atm_location l
   INNER JOIN fact_transactions t
      ON l.LocationID = t.LocationID
   GROUP BY l.State,
                    t.CardholderID),
Ranked_Customers AS (
   SELECT *,
      DENSE_RANK() OVER (PARTITION BY State
 ORDER BY Total_Transaction_Value DESC) AS Rankings
   FROM State_Customer_Transactions)
SELECT State,
            CardholderID,
            Total_Transactions,
            Total_Transaction_Value,
            Rankings
FROM Ranked_Customers
WHERE Rankings <= 3
ORDER BY State, Rankings;
```

Results | Messages

	State	CardholderID	Total_Transactions	Total_Transaction_Value	Rankings
1	Enugu	EN-008-1093	846	15325000	1
2	Enugu	EN-006-1031	445	9479000	2
3	Enugu	EN-003-1023	416	8920000	3
4	Federal Capital Territory	FC-001-1006	672	13702000	1
5	Federal Capital Territory	FC-004-1003	665	13075000	2
6	Federal Capital Territory	FC-002-1021	742	12961000	3
7	Kano	KN-005-1035	225	4992000	1
8	Kano	KN-008-1144	243	4987000	2
9	Kano	KN-002-1067	236	4963000	3
10	Lagos	LA-010-1151	208	4751000	1
11	Lagos	LA-012-1115	210	4707000	2

3. Top customers by transaction frequency

```
SELECT
   CardholderID,
   COUNT(TransactionID) AS Transaction_Frequency,
   SUM(CAST(TransactionAmount AS BIGINT)) AS Total_Transaction_Value
FROM fact_transactions
GROUP BY CardholderID
ORDER BY Transaction_Frequency DESC;
```

	CardholderID	Transaction_Frequency	Total_Transaction_Value
1	EN-008-1093	846	15325000
2	FC-002-1021	742	12961000
3	FC-004-1025	714	12173000
4	FC-001-1024	707	12374000
5	FC-002-1015	705	12422000
6	FC-005-1038	703	12953000
7	FC-006-1023	694	12723000
8	FC-004-1008	691	12410000
9	FC-005-1014	686	12506000
10	FC-002-1002	685	11927000
11	FC-001-1039	684	12709000

4. Transaction behavior by gender

```
SELECT c.Gender,
        COUNT(t.TransactionID) AS Total_Transactions,
        SUM(CAST(t.TransactionAmount AS BIGINT)) AS Total_Transaction_Value,
        ROUND(AVG(CAST(t.TransactionAmount AS DECIMAL(18,2))),
              2) AS Avg_Transaction_Value
FROM fact_transactions t
INNER JOIN customers c
   ON t.CardholderID = c.CardholderID
GROUP BY c.Gender
ORDER BY Total_Transaction_Value DESC;
```

Gender	Total_Transactions	Total_Transaction_Value	Avg_Transaction_Value
M	1072882	19298729000	17987.750000
F	1070956	19257156000	17981.280000

5. Transaction behavior by occupation

```
SELECT  c.Occupation,
```

```
        COUNT(t.TransactionID) AS Total_Transactions,
        SUM(CAST(t.TransactionAmount AS BIGINT)) AS Total_Transaction_Value,
        ROUND(
                AVG(
                CAST(t.TransactionAmount AS DECIMAL(18,2))
                ),2) AS Avg_Transaction_Value
FROM fact_transactions t
INNER JOIN customers c
ON t.CardholderID = c.CardholderID
GROUP BY c.Occupation
ORDER BY Total_Transaction_Value DESC;
```

	Occupation	Total_Transactions	Total_Transaction_Value	Avg_Transaction_Value
1	Student	261553	4733705000	18098.450000
2	Health Professional	188916	3374284000	17861.290000
3	Trader	170005	3011565000	17714.570000
4	Lawyer	163851	2945629000	17977.490000
5	Engineer	155841	2788380000	17892.470000
6	Marketing Executive	141885	2536289000	17875.670000
7	Human Resources Specialist	139773	2502688000	17905.380000
8	Manager	129951	2357493000	18141.400000
9	Accountant	130303	2353140000	18058.990000
10	Software Developer	107900	1965266000	18213.770000
11	Data Analyst	95575	1737160000	18175.880000

6. Transaction behavior by account type

```
SELECT c.AccountType,
        COUNT(t.TransactionID) AS Total_Transactions,
        SUM(CAST(t.TransactionAmount AS BIGINT)) AS Total_Transaction_Value,
        ROUND(AVG(
        CAST(t.TransactionAmount AS DECIMAL(18,2))),2) AS Avg_Transaction_Value
FROM fact_transactions t
INNER JOIN customers c
ON t.CardholderID = c.CardholderID
GROUP BY c.AccountType
ORDER BY Total_Transaction_Value DESC;
```

	AccountType	Total_Transactions	Total_Transaction_Value	Avg_Transaction_Value
1	Savings	1209361	21752485000	17986.760000
2	Current	934477	16803400000	17981.610000

7. Transaction behavior by age group (derived from BirthDate)

```
WITH Customer_Age AS (
```

```
    SELECT CardholderID,
                DATEDIFF(YEAR, Birth_Date, GETDATE()) AS Age
    FROM customers )
SELECT
    CASE
        WHEN Age < 25 THEN 'Below 25'
        WHEN Age BETWEEN 25 AND 34 THEN '25-34'
        WHEN Age BETWEEN 35 AND 44 THEN '35-44'
        WHEN Age BETWEEN 45 AND 54 THEN '45-54'
        WHEN Age BETWEEN 55 AND 64 THEN '55-64'
        ELSE '65+'
    END AS Age_Group,
    COUNT(t.TransactionID) AS Total_Transactions,
    SUM(CAST(t.TransactionAmount AS BIGINT)) AS Total_Transaction_Value,
    ROUND(AVG(CAST(t.TransactionAmount AS DECIMAL(18,2))
                ),2) AS Avg_Transaction_Value
FROM fact_transactions t
INNER JOIN Customer_Age ca
    ON t.CardholderID = ca.CardholderID
GROUP BY
    CASE
        WHEN Age < 25 THEN 'Below 25'
        WHEN Age BETWEEN 25 AND 34 THEN '25-34'
        WHEN Age BETWEEN 35 AND 44 THEN '35-44'
        WHEN Age BETWEEN 45 AND 54 THEN '45-54'
        WHEN Age BETWEEN 55 AND 64 THEN '55-64'
        ELSE '65+'
    END
ORDER BY Total_Transaction_Value DESC;
```

	Age_Group	Total_Transactions	Total_Transaction_Value	Avg_Transaction_Value
1	25-34	712520	12729377000	17865.290000
2	35-44	626248	11258945000	17978.410000
3	45-54	296455	5375337000	18132.050000
4	55-64	211824	3843050000	18142.660000
5	Below 25	194829	3493349000	17930.330000
6	65+	101962	1855827000	18201.160000

8. Wisabi Vs non-Wisabi customer transaction volume and valule

```
    SELECT CASE
                WHEN c.IsWisabi = 1 THEN 'Wisabi Customer'
                ELSE 'Non-Wisabi Customer'
```

```
            END AS Customer_Type,
            COUNT(t.TransactionID) AS Total_Transactions,
            SUM(CAST(t.TransactionAmount AS BIGINT)) AS Total_Transaction_Value,
            ROUND(AVG(CAST(t.TransactionAmount AS DECIMAL(18,2))),
                    2) AS Avg_Transaction_Value
FROM fact_transactions t
INNER JOIN customers c
  ON t.CardholderID = c.CardholderID
GROUP BY CASE
                WHEN c.IsWisabi = 1 THEN 'Wisabi Customer'
                 ELSE 'Non-Wisabi Customer'
              END
ORDER BY Total_Transaction_Value DESC;
```

	Customer_Type	Total_Transactions	Total_Transaction_Value	Avg_Transaction_Value
1	Wisabi Customer	1723398	30983585000	17978.190000
2	Non-Wisabi Customer	420440	7572300000	18010.420000

9. Average number of transactions per customer per month.

```
WITH Monthly_Customer_Transactions AS (
        SELECT CardholderID,
              YEAR(TransactionStartDateTime) AS Year_No,
              MONTH(TransactionStartDateTime) AS Month_No,
              COUNT(TransactionID) AS Monthly_Transaction_Count
        FROM fact_transactions
        GROUP BY CardholderID,
        YEAR(TransactionStartDateTime),
        MONTH(TransactionStartDateTime)   )
 SELECT
        AVG(CAST(Monthly_Transaction_Count AS FLOAT))
FROM  Monthly_Customer_Transactions;
```

	Avg_Transactions_Per_Customer_Per_Month
1	20.2577578712628

10. Distribution of customers by home ATM location (ATMID)

```
SELECT
   ATMID AS ATMID,
```

```
    COUNT(CardholderID) AS Total_Customers
FROM customers
GROUP BY ATMID
ORDER BY Total_Customers DESC;
```

	ATMID	Total_Customers
1	LA-004	323
2	LA-005	301
3	LA-008	301
4	LA-011	298
5	LA-009	295
6	LA-012	295
7	LA-015	292
8	LA-006	292
9	LA-003	291
10	LA-014	289
11	LA-001	288

	ATMID	Total_Customers
11	LA-001	288
12	LA-013	286
13	LA-010	284
14	LA-007	284
15	LA-002	283
16	KN-002	250
17	KN-006	239
18	KN-004	229
19	KN-005	220
20	KN-007	213
21	KN-010	209

	ATMID	Total_Customers
21	KN-010	209
22	KN-001	209
23	KN-008	206
24	KN-003	203
25	KN-009	193
26	RI-001	133
27	RI-005	125
28	RI-010	124
29	RI-007	121
30	RI-006	118
31	RI-003	111

ATM Location & Infrastructure

1. Number of ATMs per city and state

```
SELECT State,
       City,
       SUM(No_of_ATMs) AS Total_ATMs
FROM [atm_location ]
GROUP BY State,City
ORDER BY Total_ATMs
```

	State	City	Total_ATMs
1	Enugu	Nkanu	2
2	Enugu	Nsukka	2
3	Rivers State	Obio-Akpor	2
4	Enugu	Oji	2
5	Rivers State	Okrika	2
6	Rivers State	Opobo	2
7	Rivers State	Oyigbo	2
8	Rivers State	Ahoada	2
9	Enugu	Awgu	2
10	Rivers State	Bonny	2
11	Rivers State	Eleme	2

	State	City	Total_ATMs
9	Enugu	Awgu	2
10	Rivers State	Bonny	2
11	Rivers State	Eleme	2
12	Rivers State	Emohua	2
13	Rivers State	Gokana	2
14	Enugu	Udi	2
15	Rivers State	Port Harc...	5
16	Enugu	Enugu	11
17	Federal Ca...	Abuja	15
18	Kano	Kano	23
19	Lagos	Lagos	49

2. Revenue (transaction value) generated per location

```
SELECT t1.Location_Name,
  SUM(CAST(t2.TransactionAmount AS BIGINT)) AS Total_Value
FROM [atm_location ] as t1
LEFT JOIN fact_transactions as t2
      ON t1.LocationID = t2.LocationID
```

```
GROUP BY t1.Location_Name
ORDER BY Total_Value DESC
```

	Location_Name	Total_Value
1	Lekki ATM	1041469000
2	Surulere ATM	978496000
3	Ikorodu ATM	973858000
4	Oshodi ATM	967518000
5	Ajah ATM	961600000
6	Festac ATM	957593000
7	Yaba ATM	955450000
8	Ikeja ATM	940147000
9	Ikoyi ATM	933887000
10	Ogba ATM	930370000
11	Apapa ATM	923069000

	Location_Name	Total_Value
11	Apapa ATM	923069000
12	Victoria Island ATM	921525000
13	Sabon Gari ATM	919076000
14	Maryland ATM	918870000
15	Wisabi Bank HQ	915894000
16	Agege ATM	912502000
17	Tarauni ATM	883299000
18	Port Harcourt ATM	861191000
19	Nassarawa ATM	849566000
20	Kofar Ruwa ATM	811589000
21	Eleme ATM	807200000

	Location_Name	Total_Value
22	Opobo ATM	794208000
23	Dala ATM	785074000
24	Gokana ATM	775885000
25	Warawa ATM	765191000
26	Wisabi Bank Kan...	764045000
27	Udi ATM	763474000
28	Gwale ATM	758105000
29	Emohua ATM	750466000
30	Enugu ATM	745802000
31	Fagge ATM	741015000
32	Agbani ATM	718779000

3. Transaction volume per ATM machine at each location.

```
SELECT t1.Location_Name,
  COUNT(t2.TransactionID)  AS Total_Volume
FROM [atm_location ]  AS t1
LEFT JOIN fact_transactions AS t2
        ON t1.LocationID = t2.LocationID
GROUP BY t1.Location_Name
ORDER BY Total_Volume DESC
```

	Location_Name	Total_Volume
1	Lekki ATM	55465
2	Sabon Gari ATM	52823
3	Ikorodu ATM	51754
4	Surulere ATM	51533
5	Ajah ATM	51228
6	Oshodi ATM	51076
7	Yaba ATM	51044
8	Festac ATM	50418
9	Tarauni ATM	50381
10	Ikeja ATM	49643
11	Ikoyi ATM	49633

	Location_Name	Total_Volume
12	Apapa ATM	49254
13	Wisabi Bank HQ	49130
14	Ogba ATM	48989
15	Port Harcourt ...	48954
16	Victoria Island ...	48787
17	Maryland ATM	48694
18	Nassarawa ATM	48527
19	Agege ATM	48425
20	Kofar Ruwa A...	47044
21	Eleme ATM	45993
22	Opobo ATM	45552

	Location_Name	Total_Volume
22	Opobo ATM	45552
23	Dala ATM	44985
24	Gokana ATM	44566
25	Wisabi Bank K...	44201
26	Warawa ATM	44055
27	Udi ATM	43795
28	Emohua ATM	43365
29	Gwale ATM	43224
30	Enugu ATM	42759
31	Fagge ATM	42696
32	Agbani ATM	40921

4. Locations with the highest average transaction value.

```
SELECT t1.Location_Name,
  AVG(CAST(t2.TransactionAmount AS BIGINT)) AS Transaction_Value
FROM [atm_location ] AS t1
LEFT JOIN fact_transactions AS t2
ON t1.LocationID = t2.LocationID
GROUP BY t1.Location_Name
```

```
ORDER BY Transaction_Value DESC
```

	Location_Name	Transaction_Value
1	Festac ATM	18993
2	Ogba ATM	18991
3	Surulere ATM	18987
4	Oshodi ATM	18942
5	Ikeja ATM	18938
6	Victoria Island ATM	18888
7	Maryland ATM	18870
8	Agege ATM	18843
9	Ikorodu ATM	18817
10	Ikoyi ATM	18815
11	Lekki ATM	18777

	Location_Name	Transaction_Value
12	Ajah ATM	18770
13	Apapa ATM	18740
14	Yaba ATM	18718
15	Wisabi Bank HQ	18642
16	Garki ATM	17897
17	Lugbe ATM	17887
18	Wuse ATM	17840
19	FCT Abuja ATM	17817
20	Kubwa ATM	17809
21	Ahoada ATM	17777
22	Maitama ATM	17759

5. Locations with the highest count of unique cusotmers.

```
SELECT
   t1.LocationID,
   t1.Location_Name,
   COUNT(DISTINCT t2.CardholderID) AS Unique_Customers
FROM atm_location t1
LEFT JOIN customers t2
   ON t1.LocationID = t2.ATMID
GROUP BY t1.LocationID,
                 t1.Location_Name
ORDER BY Unique_Customers DESC;
```

	LocationID	Location_Name	Unique_Customers
1	LA-004	Lekki ATM	323
2	LA-005	Surulere ATM	301
3	LA-008	Ikorodu ATM	301
4	LA-011	Ajah ATM	298
5	LA-012	Oshodi ATM	295
6	LA-009	Yaba ATM	295
7	LA-006	Festac ATM	292
8	LA-015	Agege ATM	292
9	LA-003	Ikeja ATM	291
10	LA-014	Ikoyi ATM	289
11	LA-001	Wisabi Bank HQ	288

	LocationID	Location_Name	Unique_Customers
12	LA-013	Ogba ATM	286
13	LA-010	Maryland ATM	284
14	LA-007	Apapa ATM	284
15	LA-002	Victoria Island ...	283
16	KN-002	Sabon Gari ATM	250
17	KN-006	Tarauni ATM	239
18	KN-004	Nassarawa ATM	229
19	KN-005	Kofar Ruwa A...	220
20	KN-007	Dala ATM	213
21	KN-010	Warawa ATM	209
22	KN-001	Wisabi Bank K...	209

6. Distribution of locations by country (multi-country view)

```
SELECT
      Country,
      COUNT(LocationID) AS Total_ATM_Locations
   FROM atm_location
   GROUP BY Country
```

```
ORDER BY Total_ATM_Locations DESC;
```

	Country	Total_ATM_Locations
1	Nigeria	50

Calendar & Seasonal Summary

1. Number of holiday vs non-holiday transaction days

```
SELECT
   c.IsHoliday,
   COUNT(DISTINCT t.DateKey) AS Transaction_Days
FROM  fact_transactions t
INNER JOIN calendar c
   ON t.DateKey = c.Date
GROUP BY c.IsHoliday;
```

	IsHoliday	Transaction_Days
1	0	357
2	1	8

2. Highest single-day transaction volume

```
SELECT TOP 1
   CAST(TransactionStartDateTime AS DATE) AS Transaction_Date,
   COUNT(TransactionID) AS Total_Transactions
FROM fact_transactions
GROUP BY CAST(TransactionStartDateTime AS DATE)
ORDER BY Total_Transactions DESC;
```

	Transaction_Date	Total_Transactions
1	2022-03-03	11726

3. Monthly average daily transactions

```
SELECT YEAR(Transaction_Date) AS Year_No,
     MONTH(Transaction_Date) AS Month_No,
     AVG(Daily_Transactions) AS Avg_Daily_Transactions
FROM
      (SELECT CAST(TransactionStartDateTime AS DATE) AS Transaction_Date,
                 COUNT(TransactionID) AS Daily_Transactions
       FROM fact_transactions
```

```
        GROUP BY CAST(TransactionStartDateTime AS DATE)
      ) AS DailyData
GROUP BY YEAR(Transaction_Date),
                MONTH(Transaction_Date)
ORDER BY Year_No, Month_No;
```

	Year_No	Month_No	Avg_Daily_Transactions
1	2022	1	5771
2	2022	2	5764
3	2022	3	6333
4	2022	4	5765
5	2022	5	5932
6	2022	6	5773
7	2022	7	5947
8	2022	8	5764
9	2022	9	5781
10	2022	10	5940
11	2022	11	5744
12	2022	12	5938

4. Quarter with highest and lowest performance

```
WITH Quarterly_Transactions AS
      (SELECT DATEPART(YEAR, TransactionStartDateTime) AS Year_No,
            DATEPART(QUARTER, TransactionStartDateTime) AS Quarter_No,
            COUNT(TransactionID) AS Total_Transactions
      FROM fact_transactions
            GROUP BY DATEPART(YEAR, TransactionStartDateTime),
            DATEPART(QUARTER, TransactionStartDateTime) )
SELECT *,
```

```
 RANK()
  OVER (ORDER BY Total_Transactions DESC) AS Highest_Performance_Rank,
 RANK()
 OVER (ORDER BY Total_Transactions ASC) AS Lowest_Performance_Rank
FROM Quarterly_Transactions;
```

	Year_No	Quarter_No	Total_Transactions	Highest_Performance_Rank	Lowest_Performance_Rank
1	2022	2	530075	4	1
2	2022	3	536517	3	2
3	2022	1	536658	2	3
4	2022	4	540588	1	4

Diagnostic Analysis

1. ATM underutilization Analysis

- Which ATM locations have low transactions despite having a high number of ATMs?

```
SELECT l.Location_Name,
        l.No_of_ATMs,
        COUNT(f.TransactionID) AS Total_Transactions,
        COUNT(f.TransactionID) * 1.0 / l.No_of_ATMs AS Txn_per_ATM
FROM atm_location l
JOIN fact_transactions f
```

```
ON l.LocationID = f.LocationID
GROUP BY l.Location_Name, l.No_of_ATMs
ORDER BY Txn_per_ATM ASC;
```

	Location_Name	No_of_ATMs	Total_Transactions	Txn_per_ATM
1	FCT Abuja ATM	5	29828	5965.600000
2	Wisabi Bank HQ	7	49130	7018.571428
3	Enugu ATM	5	42759	8551.800000
4	Wisabi Bank Kano Branch ATM	5	44201	8840.200000
5	Port Harcourt ATM	5	48954	9790.800000
6	Garki ATM	2	20747	10373.500000
7	Lugbe ATM	2	23290	11645.000000
8	Wuse ATM	2	27560	13780.000000
9	Maitama ATM	2	27875	13937.500000
10	Kubwa ATM	2	30352	15176.000000
11	Agege ATM	3	48425	16141.666666

	Location_Name	No_of_ATMs	Total_Transactions	Txn_per_ATM
12	Maryland ATM	3	48694	16231.333333
13	Victoria Island ATM	3	48787	16262.333333
14	Ogba ATM	3	48989	16329.666666
15	Apapa ATM	3	49254	16418.000000
16	Ikoyi ATM	3	49633	16544.333333
17	Ikeja ATM	3	49643	16547.666666
18	Nsukka ATM	2	33168	16584.000000
19	Festac ATM	3	50418	16806.000000
20	Yaba ATM	3	51044	17014.666666
21	Oshodi ATM	3	51076	17025.333333
22	Awgu ATM	2	34074	17037.000000

- Is low usage due to fewer unique customers visiting the ATM?

```
SELECT
   l.Location_Name,
   COUNT(DISTINCT f.CardholderID) AS Unique_Customers,
   COUNT(f.TransactionID) AS Total_Transactions
FROM atm_location l
JOIN fact_transactions f
ON l.LocationID = f.LocationID
GROUP BY l.Location_Name
ORDER BY Unique_Customers ASC;
```

	Location_Name	Unique_Customers	Total_Transactions
1	Garki ATM	33	20747
2	Lugbe ATM	37	23290
3	Maitama ATM	44	27875
4	Wuse ATM	44	27560
5	FCT Abuja ATM	47	29828
6	Kubwa ATM	48	30352
7	Nsukka ATM	81	33168
8	Awgu ATM	82	34074
9	Abakpa ATM	92	37601
10	Nkanu ATM	93	38865
11	Emene ATM	94	38692

	Location_Name	Unique_Customers	Total_Transactions
12	Obio-Akpor ATM	96	35336
13	Oji River ATM	98	40376
14	Agbani ATM	99	40921
15	Bonny ATM	103	38035
16	Oyigbo ATM	103	37804
17	Enugu ATM	104	42759
18	Udi ATM	107	43795
19	Ahoada ATM	109	39881
20	Okrika ATM	111	40612
21	Emohua ATM	118	43365
22	Gokana ATM	121	44566

- Do certain transaction types dominate in low-performing locations?

```
SELECT l.Location_Name,
       t.TransactionTypeName,
       COUNT(*) AS Txn_Count
FROM fact_transactions f
JOIN atm_location l ON f.LocationID = l.LocationID
JOIN transaction_type t ON f.TransactionTypeID = t.TransactionTypeID
GROUP BY l.Location_Name, t.TransactionTypeName
```

```
ORDER BY l.Location_Name, Txn_Count DESC;
```

Results | Messages

	Location_Name	TransactionTypeName	Txn_Count
1	Abakpa ATM	Withdrawal	20958
2	Abakpa ATM	Transfer	8372
3	Abakpa ATM	Deposit	4191
4	Abakpa ATM	Balance Inquiry	4080
5	Agbani ATM	Withdrawal	22601
6	Agbani ATM	Transfer	9240
7	Agbani ATM	Balance Inquiry	4562
8	Agbani ATM	Deposit	4518
9	Agege ATM	Withdrawal	27019
10	Agege ATM	Transfer	10713
11	Agege ATM	Balance Inquiry	5476

	Location_Name	TransactionTypeName	Txn_Count
12	Agege ATM	Deposit	5217
13	Ahoada ATM	Withdrawal	22242
14	Ahoada ATM	Transfer	8880
15	Ahoada ATM	Balance Inquiry	4389
16	Ahoada ATM	Deposit	4370
17	Ajah ATM	Withdrawal	28384
18	Ajah ATM	Transfer	11394
19	Ajah ATM	Deposit	5797
20	Ajah ATM	Balance Inquiry	5653
21	Apapa ATM	Withdrawal	27386
22	Apapa ATM	Transfer	10929

- Does usage drop during specific hours of the day in these locations?

```
SELECT l.Location_Name,
        h.Hour_Start_Time,
        COUNT(*) AS Txn_Count
    FROM fact_transactions f
    JOIN atm_location l ON f.LocationID = l.LocationID
    JOIN hours_table h ON f.Hour_Key = h.Hour_Key
    GROUP BY l.Location_Name, h.Hour_Start_Time
    ORDER BY Txn_Count ASC;
```

	Location_Name	Hour_Start_Time	Txn_Count
1	Ajah ATM	02:00:00.0000000	1
2	Festac ATM	02:00:00.0000000	1
3	Ikeja ATM	02:00:00.0000000	1
4	Oshodi ATM	02:00:00.0000000	1
5	Yaba ATM	02:00:00.0000000	1
6	Ogba ATM	03:00:00.0000000	1
7	Oshodi ATM	03:00:00.0000000	1
8	Wisabi Bank HQ	03:00:00.0000000	1
9	Ikeja ATM	00:00:00.0000000	1
10	Lekki ATM	00:00:00.0000000	1
11	Agege ATM	01:00:00.0000000	1

	Location_Name	Hour_Start_Time	Txn_Count
12	Surulere ATM	01:00:00.0000000	1
13	Maryland ATM	04:00:00.0000000	1
14	Victoria Island ...	01:00:00.0000000	2
15	Victoria Island ...	00:00:00.0000000	2
16	Warawa ATM	04:00:00.0000000	108
17	Victoria Island ...	06:00:00.0000000	109
18	Ungogo ATM	06:00:00.0000000	113
19	Wisabi Bank HQ	05:00:00.0000000	115
20	Gwale ATM	05:00:00.0000000	116
21	Maryland ATM	05:00:00.0000000	119
22	Gwale ATM	06:00:00.0000000	120

- Are these ATMs used less during holidays or weekends?

```
SELECT  l.Location_Name,
            c.IsHoliday,
        c.Day_Name,
        COUNT(*) AS Txn_Count
FROM fact_transactions f
JOIN atm_location l ON f.LocationID = l.LocationID
JOIN calendar c ON f.DateKey = c.Date
GROUP BY l.Location_Name, c.IsHoliday, c.Day_Name
```

```
ORDER BY Txn_Count ASC;
```

	Location_Name	IsHoliday	Day_Name	Txn_Count
1	Garki ATM	1	Saturday	41
2	Garki ATM	1	Wednesday	53
3	Lugbe ATM	1	Wednesday	57
4	Wuse ATM	1	Saturday	62
5	Wuse ATM	1	Wednesday	70
6	Kubwa ATM	1	Wednesday	70
7	Lugbe ATM	1	Saturday	71
8	FCT Abuja ATM	1	Wednesday	74
9	Maitama ATM	1	Wednesday	75
10	Maitama ATM	1	Saturday	78
11	Awgu ATM	1	Wednesday	78
12	Obio-Akpor ATM	1	Saturday	81
13	Emene ATM	1	Saturday	82
14	FCT Abuja ATM	1	Saturday	82
15	Bonny ATM	1	Wednesday	87
16	Kubwa ATM	1	Saturday	89
17	Abakpa ATM	1	Wednesday	92
18	Awgu ATM	1	Saturday	95
19	Fagge ATM	1	Saturday	96
20	Obio-Akpor ATM	1	Wednesday	97
21	Nsukka ATM	1	Wednesday	98
22	Bonny ATM	1	Saturday	100

2. Customer Segment Impact on ATM usage

- Do customers with certain Account Types perform fewer ATM transactions?

```
SELECT c.AccountType,
        COUNT(f.TransactionID) AS Txn_Count
FROM fact_transactions f
JOIN customers c
ON f.CardholderID = c.CardholderID
GROUP BY c.AccountType
ORDER BY Txn_Count ASC;
```

	AccountType	Txn_Count
1	Current	934477
2	Savings	1209361

Is ATM usage lower among specific occupations

```
SELECT c.Occupation,
        COUNT(*) AS Txn_Count
FROM fact_transactions f
JOIN customers c
ON f.CardholderID = c.CardholderID
GROUP BY c.Occupation
ORDER BY Txn_Count ASC;
```

	Occupation	Txn_Count
1	Plumber	44402
2	Carpenter	46803
3	Welder	48740
4	Mechanic	50735
5	Truck Driver	51431
6	Farmer	52113
7	Electrician	52161
8	Construction Worker	54319
9	Factory Worker	57581
10	Data Analyst	95575
11	Software Developer	107900

	Occupation	Txn_Count
10	Data Analyst	95575
11	Software Developer	107900
12	Manager	129951
13	Accountant	130303
14	Human Resources...	139773
15	Marketing Executive	141885
16	Engineer	155841
17	Lawyer	163851
18	Trader	170005
19	Health Professional	188916
20	Student	261553

- Do Wisabi customers prefer certain ATM locations more than others?

```
SELECT    l.Location_Name,
          COUNT(*) AS Wisabi_Txn
FROM fact_transactions f
JOIN customers c ON f.CardholderID = c.CardholderID
JOIN atm_location l ON f.LocationID = l.LocationID
WHERE c.IsWisabi = 1
GROUP BY l.Location_Name
ORDER BY Wisabi_Txn DESC;
```

	Location_Name	Wisabi_Txn
1	Sabon Gari ATM	45474
2	Lekki ATM	44186
3	Ikorodu ATM	42565
4	Nassarawa ATM	41584
5	Yaba ATM	41463
6	Oshodi ATM	41400
7	Ikoyi ATM	41396
8	Surulere ATM	41283
9	Maryland ATM	40639
10	Port Harcourt ATM	40483
11	Ikeja ATM	40259

	Location_Name	Wisabi_Txn
12	Festac ATM	40025
13	Tarauni ATM	39882
14	Ajah ATM	39786
15	Apapa ATM	39062
16	Kofar Ruwa ATM	38945
17	Ogba ATM	38923
18	Victoria Island A...	38566
19	Wisabi Bank HQ	38238
20	Agege ATM	37981
21	Gokana ATM	37234
22	Emohua ATM	36757

	Location_Name	Wisabi_Txn
23	Opobo ATM	36574
24	Eleme ATM	36430
25	Warawa ATM	36052
26	Udi ATM	35721
27	Gwale ATM	35673
28	Enugu ATM	34992
29	Wisabi Bank Kan...	34567
30	Okrika ATM	33711
31	Dala ATM	33560
32	Nkanu ATM	33146
33	Agbani ATM	33077

- Which customer segments contribute least to ATM usage

```
SELECT
    c.AccountType,
    c.Occupation,
    COUNT(*) AS Txn_Count
FROM fact_transactions f
```

```
JOIN customers c
ON f.CardholderID = c.CardholderID
GROUP BY c.AccountType, c.Occupation
ORDER BY Txn_Count ASC;
```

	AccountType	Occupation	Txn_Count
1	Savings	Plumber	21380
2	Current	Welder	21782
3	Savings	Carpenter	21952
4	Current	Plumber	23022
5	Savings	Electrician	23930
6	Savings	Farmer	24300
7	Current	Carpenter	24851
8	Savings	Mechanic	25075
9	Savings	Truck Driver	25627
10	Current	Mechanic	25660
11	Current	Truck Driver	25804
12	Current	Factory Worker	26404
13	Current	Construction ...	26478
14	Savings	Welder	26958
15	Current	Farmer	27813
16	Savings	Construction ...	27841
17	Current	Electrician	28231
18	Savings	Factory Worker	31177
19	Current	Data Analyst	46095
20	Savings	Data Analyst	49480
21	Current	Software Dev...	50520
22	Savings	Software Dev...	57380

Holiday transaction behavior analysis

1. Which transactoin types increase during holidays?

```
SELECT t.TransactionTypeName,
        COUNT(*) AS Holiday_Txn
FROM fact_transactions f
JOIN calendar c ON f.DateKey = c.Date
JOIN transaction_type t ON f.TransactionTypeID = t.TransactionTypeID
WHERE c.IsHoliday = 1
GROUP BY t.TransactionTypeName;
```

	TransactionTypeName	Holiday_Txn
1	Balance Inquiry	5121
2	Deposit	4974
3	Transfer	10111
4	Withdrawal	25555

2. So wisabi customers transact more during holidays than non-wisabi users?

```
SELECT c.IsWisabi,
        COUNT(*) AS Txn_Count
FROM fact_transactions f
JOIN customers c ON f.CardholderID = c.CardholderID
JOIN calendar cal ON f.DateKey = cal.Date
```

```
WHERE cal.IsHoliday = 1
GROUP BY c.IsWisabi;
```

Results | Messages

	IsWisabi	Txn_Count
1	0	9015
2	1	36746

3. Which locations experience holiday transactioon spikes?

```
SELECT
   l.Location_Name,
   COUNT(*) AS Holiday_Txn
FROM fact_transactions f
JOIN calendar c ON f.DateKey = c.Date
JOIN atm_location l ON f.LocationID = l.LocationID
WHERE c.IsHoliday = 1
GROUP BY l.Location_Name
ORDER BY Holiday_Txn DESC;
```

	Location_Name	Holiday_Txn
1	Lekki ATM	1212
2	Ajah ATM	1114
3	Sabon Gari ATM	1109
4	Ikeja ATM	1097
5	Ikorodu ATM	1096
6	Wisabi Bank HQ	1088
7	Ikoyi ATM	1087
8	Surulere ATM	1077
9	Tarauni ATM	1068
10	Yaba ATM	1068
11	Port Harcourt ATM	1064

	Location_Name	Holiday_Txn
12	Oshodi ATM	1061
13	Ogba ATM	1029
14	Festac ATM	1027
15	Kofar Ruwa ATM	1024
16	Victoria Island A...	1022
17	Apapa ATM	1021
18	Maryland ATM	1020
19	Nassarawa ATM	1015
20	Agege ATM	1013
21	Eleme ATM	998
22	Dala ATM	982

	Location_Name	Holiday_Txn
23	Udi ATM	979
24	Opobo ATM	973
25	Gokana ATM	954
26	Gwale ATM	953
27	Wisabi Bank Kan...	950
28	Emohua ATM	924
29	Warawa ATM	911
30	Fagge ATM	903
31	Okrika ATM	893
32	Agbani ATM	880
33	Ahoada ATM	876

4. Are transactions during holidays concentrated in certain customer segments?

```
SELECT c.AccountType,
       COUNT(*) AS Txn_Count
FROM fact_transactions f
JOIN customers c ON f.CardholderID = c.CardholderID
JOIN calendar cal ON f.DateKey = cal.Date
WHERE cal.IsHoliday = 1
```

```
GROUP BY c.AccountType;
```

	AccountType	Txn_Count
1	Savings	25951
2	Current	19810

3. State-wise ATM demand variation analysis

1. Why do some states show higher ATM usage than others?

```
SELECT l.State,
        COUNT(*) AS Txn_Count
FROM fact_transactions f
JOIN atm_location l ON f.LocationID = l.LocationID
GROUP BY l.State
ORDER BY Txn_Count DESC;
```

	State	Txn_Count
1	Lagos	755073
2	Kano	458764
3	Rivers State	420098
4	Enugu	350251
5	Federal Capital Territory	159652

2. Are certain states more withdrawal-heavy

```
SELECT l.State,
        COUNT(*) AS Withdrawal_Txn
FROM fact_transactions f
JOIN atm_location l ON f.LocationID = l.LocationID
JOIN transaction_type t ON f.TransactionTypeID = t.TransactionTypeID
WHERE t.TransactionTypeName = 'Withdrawal'
GROUP BY l.State;
```

	State	Withdrawal_Txn
1	Enugu	194440
2	Federal Capital Territory	88136
3	Kano	253796
4	Lagos	419807
5	Rivers State	233672

3. Is weekend usage higher in specific states?

```
SELECT l.State,
        COUNT(*) AS Weekend_Txn
FROM fact_transactions f
JOIN atm_location l ON f.LocationID = l.LocationID
JOIN calendar c ON f.DateKey = c.Date
WHERE c.Day_of_Week IN (6,7)
GROUP BY l.State;
```

	State	Weekend_Txn
1	Enugu	49895
2	Federal Capital Territory	22749
3	Kano	65203
4	Lagos	107424
5	Rivers State	59952

4. Do holiday transactions increase more in some states?

```
SELECT l.State,
        COUNT(*) AS Holiday_Txn
FROM fact_transactions f
JOIN atm_location l ON f.LocationID = l.LocationID
JOIN calendar c ON f.DateKey = c.Date
WHERE c.IsHoliday = 1
GROUP BY l.State;
```

Results | Messages

	State	Holiday_Txn
1	Enugu	7517
2	Federal Capital Territory	3418
3	Kano	9758
4	Lagos	16032
5	Rivers State	9036

4. Transaction duration root cause analysis

1. Why do certaiin locations show longer average transaction durations?

```
SELECT c.Occupation,
        AVG(DATEDIFF(SECOND,
                f.TransactionStartDateTime,
                f.TransactionEndDateTime)) AS Avg_Duration
FROM fact_transactions f
JOIN customers c ON f.CardholderID = c.CardholderID
GROUP BY c.Occupation;
```

	Location_Name	Avg_Duration
1	Abakpa ATM	220
2	Agbani ATM	220
3	Agege ATM	249
4	Ahoada ATM	220
5	Ajah ATM	250
6	Apapa ATM	250
7	Awgu ATM	219
8	Bonny ATM	220
9	Dala ATM	292
10	Eleme ATM	219
11	Emene ATM	220

	Location_Name	Avg_Duration
12	Emohua ATM	219
13	Enugu ATM	219
14	Fagge ATM	293
15	FCT Abuja ATM	251
16	Festac ATM	250
17	Garki ATM	250
18	Gokana ATM	220
19	Gwale ATM	294
20	Ikeja ATM	250
21	Ikorodu ATM	249
22	Ikoyi ATM	250

	Location_Name	Avg_Duration
23	Kofar Ruwa A...	293
24	Kubwa ATM	249
25	Lekki ATM	249
26	Lugbe ATM	251
27	Maitama ATM	250
28	Maryland ATM	249
29	Nassarawa A...	292
30	Nkanu ATM	219
31	Nsukka ATM	219
32	Obio-Akpor A...	219
33	Ogba ATM	250

2. Is transaction duration higher during peak hours?

```
SELECT h.Hour_Start_Time,
        AVG(DATEDIFF(SECOND,
                f.TransactionStartDateTime,
                f.TransactionEndDateTime)) AS Avg_Duration
FROM fact_transactions f
JOIN hours_table h ON f.Hour_Key = h.Hour_Key
GROUP BY h.Hour_Start_Time;
```

	Hour_Start_Time	Avg_Duration
1	00:00:00.0000000	165
2	01:00:00.0000000	240
3	02:00:00.0000000	276
4	03:00:00.0000000	280
5	04:00:00.0000000	268
6	05:00:00.0000000	257
7	06:00:00.0000000	259
8	07:00:00.0000000	248
9	08:00:00.0000000	248
10	09:00:00.0000000	247
11	10:00:00.0000000	248

	Hour_Start_Time	Avg_Duration
12	11:00:00.0000000	255
13	12:00:00.0000000	254
14	13:00:00.0000000	255
15	14:00:00.0000000	255
16	15:00:00.0000000	249
17	16:00:00.0000000	245
18	17:00:00.0000000	244
19	18:00:00.0000000	235
20	19:00:00.0000000	241
21	20:00:00.0000000	253
22	21:00:00.0000000	253

17	16:00:00.0000000	245
18	17:00:00.0000000	244
19	18:00:00.0000000	235
20	19:00:00.0000000	241
21	20:00:00.0000000	253
22	21:00:00.0000000	253
23	22:00:00.0000000	252
24	23:00:00.0000000	252

3. Do transactions take longer during holidays?

```
SELECT cal.IsHoliday,
                    AVG(DATEDIFF(SECOND,
                                 f.TransactionStartDateTime,
                                 f.TransactionEndDateTime)) AS Avg_Duration_In_Seconds
        FROM fact_transactions f
        JOIN calendar cal ON f.DateKey = cal.Date
        GROUP BY cal.IsHoliday;
```

	IsHoliday	Avg_Duration_In_Seconds
1	0	248
2	1	248

4. Do specific customer occupations take more time to transact

```
SELECT c.Occupation,
          AVG(DATEDIFF(SECOND,
                       f.TransactionStartDateTime,
                       f.TransactionEndDateTime)) AS Avg_Duration
FROM fact_transactions f
JOIN customers c ON f.CardholderID = c.CardholderID
GROUP BY c.Occupation;
```

	Occupation	Avg_Duration
1	Electrician	246
2	Health Professional	256
3	Welder	246
4	Marketing Executive	253
5	Plumber	248
6	Carpenter	251
7	Lawyer	251
8	Construction Worker	247
9	Farmer	251
10	Accountant	253
11	Mechanic	248

	Occupation	Avg_Duration
10	Accountant	253
11	Mechanic	248
12	Manager	241
13	Engineer	250
14	Software Developer	237
15	Data Analyst	235
16	Student	241
17	Truck Driver	248
18	Factory Worker	251
19	Trader	256
20	Human Resources...	251

5. Wisabi vs Non-Wisabi customer behavior analysis

1. Do wisabi customers prefer certain transaction types

```
SELECT c.IsWisabi,
        t.TransactionTypeName,
        COUNT(*) AS Txn_Count
FROM fact_transactions f
JOIN customers c ON f.CardholderID = c.CardholderID
JOIN transaction_type t ON f.TransactionTypeID = t.TransactionTypeID
GROUP BY c.IsWisabi, t.TransactionTypeName;
```

	IsWisabi	TransactionTypeName	Txn_Count
1	0	Balance Inquiry	46715
2	1	Balance Inquiry	192550
3	0	Deposit	46924
4	1	Deposit	191926
5	0	Transfer	93123
6	1	Transfer	382749
7	0	Withdrawal	233678
8	1	Withdrawal	956173

2. Do wisabi users transact more during weekdays or weekends

```
SELECT c.IsWisabi,
        CASE WHEN cal.Day_of_Week IN (6,7) THEN 'Weekend'
            ELSE 'Weekday'
        END AS Day_Type,
        COUNT(*) AS Txn_Count
FROM fact_transactions f
JOIN customers c ON f.CardholderID = c.CardholderID
JOIN calendar cal ON f.DateKey = cal.Date
```

```
GROUP BY c.IsWisabi,
        CASE WHEN cal.Day_of_Week IN (6,7) THEN 'Weekend'
             ELSE 'Weekday'
        END;
```

	IsWisabi	Day_Type	Txn_Count
1	0	Weekday	360572
2	1	Weekday	1478043
3	0	Weekend	59868
4	1	Weekend	245355

3. Are wisabi users more active during specific hours?

```
SELECT  c.IsWisabi,
        h.Hour_Start_Time,
        COUNT(*) AS Txn_Count
FROM fact_transactions f
JOIN customers c ON f.CardholderID = c.CardholderID
JOIN hours_table h ON f.Hour_Key = h.Hour_Key
GROUP BY c.IsWisabi, h.Hour_Start_Time;
```

	IsWisabi	Hour_Start_Time	Txn_Count
1	0	00:00:00.0000000	2
2	1	00:00:00.0000000	2
3	0	01:00:00.0000000	1
4	1	01:00:00.0000000	3
5	0	02:00:00.0000000	3
6	1	02:00:00.0000000	2
7	0	03:00:00.0000000	2
8	1	03:00:00.0000000	1
9	0	04:00:00.0000000	274
10	1	04:00:00.0000000	1059
11	0	05:00:00.0000000	917

	IsWisabi	Hour_Start_Time	Txn_Count
12	1	05:00:00.0000000	3590
13	0	06:00:00.0000000	882
14	1	06:00:00.0000000	3776
15	0	07:00:00.0000000	22771
16	1	07:00:00.0000000	93132
17	0	08:00:00.0000000	22377
18	1	08:00:00.0000000	91936
19	0	09:00:00.0000000	22644
20	1	09:00:00.0000000	92286
21	0	10:00:00.0000000	22497
22	1	10:00:00.0000000	92154

4. Which locations are dominated by Wisabi users

```
SELECT l.Location_Name,
        COUNT(*) AS Wisabi_Txn
FROM fact_transactions f
JOIN customers c ON f.CardholderID = c.CardholderID
JOIN atm_location l ON f.LocationID = l.LocationID
WHERE c.IsWisabi = 1
GROUP BY l.Location_Name
ORDER BY Wisabi_Txn DESC;
```

	Location_Name	Wisabi_Txn
1	Sabon Gari ATM	45474
2	Lekki ATM	44186
3	Ikorodu ATM	42565
4	Nassarawa ATM	41584
5	Yaba ATM	41463
6	Oshodi ATM	41400
7	Ikoyi ATM	41396
8	Surulere ATM	41283
9	Maryland ATM	40639
10	Port Harcourt ATM	40483
11	Ikeja ATM	40259

	Location_Name	Wisabi_Txn
12	Festac ATM	40025
13	Tarauni ATM	39882
14	Ajah ATM	39786
15	Apapa ATM	39062
16	Kofar Ruwa ATM	38945
17	Ogba ATM	38923
18	Victoria Island A...	38566
19	Wisabi Bank HQ	38238
20	Agege ATM	37981
21	Gokana ATM	37234
22	Emohua ATM	36757

	Location_Name	Wisabi_Txn
23	Opobo ATM	36574
24	Eleme ATM	36430
25	Warawa ATM	36052
26	Udi ATM	35721
27	Gwale ATM	35673
28	Enugu ATM	34992
29	Wisabi Bank Kan...	34567
30	Okrika ATM	33711
31	Dala ATM	33560
32	Nkanu ATM	33146
33	Agbani ATM	33077